

Tirthajit Baruah

PhD in Cyber Physical Systems, IISc
BS-MS in Electrical Engineering and Computer Science
(Minor in Data Science and Engineering), IISER Bhopal

+91-9127861598
tirthajitb@iisc.ac.in
GitHub Google Scholar

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
PhD	Indian Institute of Science, Bangalore	–	2023-Present
Bachelor of Science - Master of Science	Indian Institute of Science Education and Research, Bhopal	8.3/10	2018-2023

PUBLICATIONS

1. **T. Baruah**, K. Jamadar, P. Rathore, "*Two-Fold Patch Perturbation for Efficient Self-Supervised Learning in 3D Medical Imaging*" Accepted at the International Joint Conference on Artificial Intelligence (IJCAI-ECAI), Bremen, 2026. [LINK]
2. **T. Baruah**, P. Rathore, "*Translating Classifier Scores into Clinical Impact: Calibrated Risk and Queueing Simulation for AI-Assisted Radiology Worklist Triage*" Accepted at DAI & AIMedHealth @ Association for the Advancement of Artificial Intelligence (AAAI), Singapore, 2026. [LINK]
3. A. Mazumder, C. Garg, **T. Baruah**, P. Rathore, "*DIME: Deterministic Information Maximizing Autoencoder*" Accepted at DeLTa @ International Conference on Learning Representations (ICLR), Singapore, 2025. [LINK]
4. A. Mazumder, **T. Baruah**, B. Kumar, R. Sharma, V. Pattanaik, P. Rathore, "*Learning Low-Rank Latent Spaces with Simple Deterministic Autoencoder: Theoretical and Empirical Insights*," Accepted at IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), Hawaii, 2024. [LINK]
5. A. Mazumder, **T. Baruah**, A. Singh, P. Krishna, V. Pattanaik, P. Rathore, "*DeepVAT: A Self-Supervised Technique for Cluster Assessment for Image Datasets*," Accepted at ViPrior @ IEEE/CVF International Conference on Computer Vision (ICCV), Paris, 2023. [LINK]
6. H. Wasnik, R. Gandhi, N. Patil, R. Behera, A. Golait, T. Patel, & **T. Baruah**, "*A Comprehensive Review of Molecular Biology and Genetics of Cataract*," International Journal for Research in Applied Sciences and Biotechnology, 8(2), 11-20. [LINK]

KEY PROJECTS

- **Internship Project** Jun 2023 - Jul 2023
Supervisor: Dr. Punit Rathore CPS, IISc Bangalore (Project Page)
 - Formulated a novel autoencoder (AE) architecture that gives low-rank latent space.
 - This architecture is then used for various downstream tasks like classification, clustering, and generative tasks, and it outperforms the baseline and SOTA autoencoders including generative AE's like VAE and Flow-based AE.
- **Master of Science Thesis Project** Aug 2022 - Apr 2023
Supervisor: Dr. Prathosh AP ECE, IISc Bangalore
 - **Representation Learning for Electronic Healthcare Records**
 - Develop augmentation techniques for Electronic Healthcare Records
 - Learn time series representations using self-supervised modeling
 - Handling large-scale real-world electronic health record (MIMIC III, Physionet 2012)
- **Bachelor of Science Final Project** Dec 2021 - Apr 2022
Supervisor: Dr. Haroon R Lone EECS, IISER Bhopal
 - **Calibrating Attention of Students in Classroom**
 - Studying the already used techniques and sensors (Galvanic, Electroencephalography, Visual)
 - Deploy Raspberry Pi module and camera to collect student data from the classroom
 - Train a real-time attentiveness detection model using the collected data and computer vision techniques
- **Summer Project: Water Rediscover** Apr - Jul 2013
Project Instructor: Dr. Achintya N. Bezbaruah North Dakota State University
 - Developing wastewater management model for recycling them into potable water
 - Utilizing locally available techniques and plants (with cleansing property)

EXPERIENCE

- **Dept. of Cyber Physical Systems** *Jan - May 2025*
IISc Bengaluru
Teaching Assistant: DA218o (Probabilistic Machine Learning: Theory and Applications)
 - Conducting weekly programming tutorials for the topics taught in class
 - Assisting the instructor in formulating and evaluating assignments and exams
- **Dept. of Data Science and Engineering** *Aug - Dec 2022*
IISER Bhopal
Teaching Assistant: DSE317/617 (Machine Learning)
 - Delivering lectures on supplementary topics of the course
 - Mentoring and guiding students (Batch 2020/19) throughout the course
 - Assisting the instructor in formulating and evaluating assignments and exams
- **Dept. of Electrical Engineering & Computer Science** *Apr - May 2022*
IISER Bhopal
Teaching Assistant: ECS102 (Introduction to Programming)
 - Mentoring and guiding students (Batch 2021) throughout the course
 - Conducting lab sessions for assisting the students with their difficulty in assignments and course
- **Artificial Intelligence and Cyber Physical System for Agriculture @ iHub - AWaDH** *May - Jul 2021*
IIT Ropar
Workshop, Instructor: Dr. Neeraj Goel
 - Collecting agricultural data via sensors, deploying them to AI model training
 - Android app development for agricultural purposes
 - Capstone project: **“Real-time Pest Detection using MobileNet-SSD & Farm Monitoring”**

TECHNICAL SKILLS

- **Programming:** Python (PyTorch), Distributed Data Parallel, Linux
- **Hardware Tools:** ESP32, Arduino, Raspberry Pi, IoT sensors

RELEVANT COURSES

- **Mathematics:** Stochastic Modelling & Applications, Linear Algebra, Probability & Statistics, Calculus
- **Computer Science:** Data Structure & Algorithm, Discrete Maths, Internet of Things
- **Data Science:** Pattern Recognition & Neural Networks, Theory and Applications of Bayesian Learning, AI for Medical Image Analysis, Digital Image Processing
- **Biology:** Bio-molecules, Cell Biology, Biostatistics, Bio-informatics
- **Electrical and Electronics:** Signals & Systems, Principles of Communication, Electronic Devices, Electromagnetic Theory, Smart Sensing Technologies

LANGUAGES

- **English**, Full Professional Proficiency (**TOEFL-iBT score: 101/120**) *Oct 2022*
- **Hindi**, Native/Bilingual Proficiency
- **Assamese**, Native/Bilingual Proficiency

POSITIONS OF RESPONSIBILITY

- **Coordinator**, Karate Club, IISER Bhopal *2018 - 2021*
- **Prefect**, Students Council, Army Public School, Tezpur *2016*

EXTRA-CURRICULARS

- **Gold Medal:** Athletics 4*100m relay, Inter IISER Sports Meet, 2022
 - **Visharad in Tabla:** 5-years classical training of tabla instrument
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